CIS352 Intro to Enterprise Computing

Chapter 1 Assignment

Make the following changes to Chapter's 1 CALC2000 program

1) Add program header documentation

```
identification division.

program-id. calc2000.
*  Programmer.: Debbie Johnson
*  Date.......: 2026.01.26
*  GitHub URL.:
*  https://github.com/dejohns2/COBOL-Chapter1-Assignment.git
*  Description: This program calculates future values for an
*  investment and doubles the investment amount twice.

environment division.
```

2) Instead of using the ACCEPT statement for accepting the user's input,
   rename the **user-entries** to be **input-values**, plus assign initial values for all input fields.
   - Investment-amount value 1000
   - Number-of-years value 10
   - Yearly-interest-rate value 5.5

3) Delete the edited-future-value data item and replace it with the following two data items that will be used for displaying numeric edit items:

```
01  work-fields.

    05  future-value                pic 9(7)v99.
    05  year-counter                pic 99.

    05  edited-whole-value          pic zz,zzz,zz9.
    05  edited-decimal-value        pic zzz,zzz.99.
```

4) Remove the unused 110-get-user-values paragraph
   Plus, remove the following logic for getting the user's initial input

```
100-calculate-future-value.

    display "----------------------------------------".
    display "To end the program, enter 0.".
    display "To perform another calculation, enter 1.".
    accept  number-entered.
    display "----------------------------------------".
    if number-entered = 1
        perform 110-get-user-values
```

NOTE: Don't forget to unindent the code that used to be inside the if statement so it's at the start

of Column B.

5) Convert the **000-calculate-future-values** paragraph to do the following:
   a. Display the following program header output

```
Calculating Future Values
```

   b. Call perform 100-calculate-future-value without using an until clause
   c. Then double the investment-amount using a COMPUTE statement and call the 100 paragraph again
   d. Plus, it will do this a third time, too

6) Add a new paragraph called **140-display-values** that will be called at the end of **100-calculate-future-value** to produce the following output:

```
-----------------------------
Investment Amount =      1,000
Number of Years   =         10
Yearly Interest   =       5.50
Future value      =   1,708.16
```

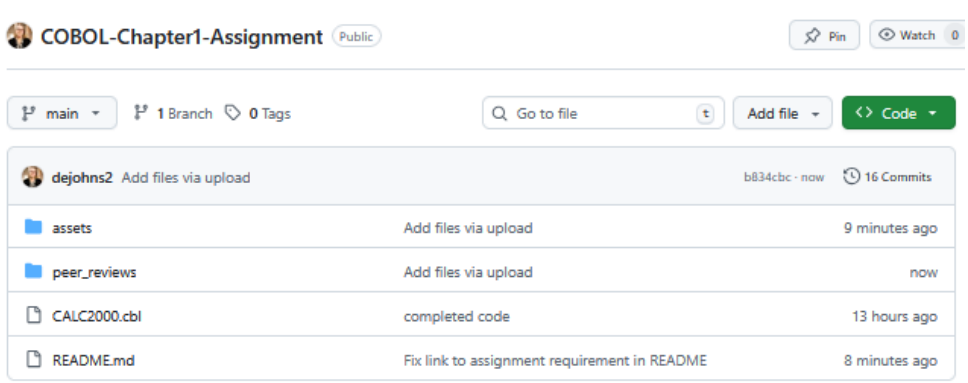
Additional requirements:

1) Your goal is to NOT use AI for this assignment, but if you do, use it sparely. If you use AI, you need to document where you used it in your code and justify your reason. I'm okay with you using it to get help, for very specific items, but not to write your code for you.

2) Well documented code.
   a. https://google.github.io/styleguide/jsguide.html
   b. https://gomakethings.com/whats-the-best-way-to-document-javascript/

3) Must use GitHub
   a. Invite me to your repository (dejohns2)
   b. Numerous commits which will prove who did work
   c. Balance work across code and README

Repository Structure Example:



d. Professional README.md
   https://github.com/banesullivan/README
   https://github.com/matiassingers/awesome-readme?tab=readme-ov-file
   i. Fully explain what the program does
   ii. Embedded screenshots (stored in a subfolder)
   iii. Bullet point list of what new concepts were used
   iv. links to the author's profiles

README.md Example:

CALC2000 (COBOL) — Future Value Investment Calculator

A small COBOL program that calculates the future value of an investment using a fixed annual interest rate over a fixed number of years. After the first calculation, it **doubles the investment amount twice**, recalculating the future value each time (so you get **three** future-value results total).

What it does

For each run, the program:

1. Calculates the future value for the initial investment.
2. Doubles the investment amount and calculates again.
3. Doubles the investment amount again and calculates a third time.

The future value is computed iteratively (year-by-year) as:

Marist > JOB07764 > KC02606.JOB07764.STEP1.GO.SYSOUT.106

```
  Calculating Future Values
  -----------------------------
  Investment Amount =      1,000
  Number of Years   =         10
  Yearly Interest   =       5.50
  Future value      =   1,708.16

  -----------------------------
  Investment Amount =      2,000
  Number of Years   =         10
  Yearly Interest   =       5.50
  Future value      =   3,416.29

  -----------------------------
  Investment Amount =      4,000
  Number of Years   =         10
  Yearly Interest   =       5.50
  Future value      =   6,832.58

  End of session.
```

For the full program details see Assignment Requirement

COBOL Concepts covered in this assignment were:

1. Program header level documentation
2. Defining working storage data items
3. Declaring Procedure Division paragraphs
4. Moving and computing values
5. Displaying output

Authors

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How to embed an image:

```
![Program Output](assets/CALC2000_Output.png)
```

How to link to an image:

```
1. [Program header level documentation](assets/program_header.png)
```

4) PDF Peer review in a subfolder. Use the other person's name as the file name. A peer review should focus on the person's strengths and possibly providing helpful suggestions for future group work.

   To give a good peer review on a programming assignment, be constructive, focus on learning, and use a positive, clear tone, starting with praise and then suggestions for improvement. Frame feedback as questions or requests, provide code examples, and avoid nitpicking minor issues, ensuring the review is a collaborative effort to improve future code and collaboration.

5) Attach your program file to the assignment along with your Good, Bad, and Ugly (GBU) reflection written at a college level. IMPORTANT: Include, your GitHub URL in this document.